

HOGAN NGUYEN

469-720-5180 | hoganngu756@gmail.com | linkedin.com/in/hogan-nguyen | github.com/hoganngu756 | hoganngu756.github.io

EDUCATION

University of Texas at Dallas
Master of Science in Computer Science

August 2025 – May 2027 (Expected)
Richardson, TX

University of Texas at Dallas
Bachelor of Science in Software Engineering

December 2024
Richardson, TX

RELEVANT EXPERIENCE

Software Infrastructure Engineer Intern
American Heart Association

Incoming September 2026
Dallas, TX

- Joining the infrastructure engineering team to support cloud and platform infrastructure

AI Consultant Intern
IBM

May 2026 – August 2026
Austin, TX

- Built the document-analysis module for a due diligence agent in Python, pairing LangChain deepagents with Azure OpenAI and Foundry vector search to extract deal terms, replacing a multi-hour manual review per deal
- Created a reporting dashboard using React, Next.js, TypeScript, and PostgreSQL to automate weekly client deliverables, eliminating manual PowerPoint workflows and reducing prep time by 40%
- Deployed the team's first internal AI agent to IBM Cloud using Terraform, authoring IaC modules for provisioning and environment configuration
- Translated client business requirements into technical specs during sprint planning and stand-ups, tracking deliverables in monday.com for real-time engagement visibility

Problem Design Engineer
Idler (YC25)

January 2026 – Present
Remote (Part-time)

- Authored edge-case test suites and complex coding problems in Python and Java to benchmark LLM code-generation accuracy and identify subtle logic vulnerabilities
- Reviewed 50+ code submissions across OOP and scripting stacks to maintain quality standards and test coverage prior to benchmark deployment

Software Engineer Intern
Blackhawk Network

June 2024 – August 2024
Coppell, TX

- Automated incident ticket escalation workflows in Python on ServiceNow, reducing manual intervention and cutting average response time by 20%
- Constructed custom Splunk and New Relic dashboards correlating application logs with system metrics into a single view of service health, surfacing infrastructure bottlenecks previously caught only after tickets were filed
- Authored runbooks and standardized resolution procedures across 100+ incident tickets in ServiceNow, giving the on-call rotation consistent first-response triage steps

PROJECTS

Document Security and Prompt Scanner | *Java, React, Spring Boot, Gemini API, SQLite*

[GitHub](#)

- Architected a four-layer security engine using regex heuristics, PDF structure analysis, visual-obfuscation auditing, and Gemini LLM API to detect hidden prompt injections and jailbreaks across PDF text and embedded images, covering 25+ distinct injection patterns
- Achieved a 90% detection rate with under 8% false positives against a 150-document adversarial test corpus, surfacing flagged results through a full-stack upload and scanning interface

Kubernetes Resource and Cost Monitor | *Python, Kubernetes, Prometheus, OpenCost, Docker, MCP*

[GitHub](#)

- Developed a pipeline that cross-references Prometheus container metrics against pricing models and visualizes request-versus-usage deltas, flagging 30% of workloads in a 40-pod cluster as over-provisioned
- Exposed the pipeline as an MCP server, letting LLM agents query live cluster utilization and cost data through structured tool calls for rightsizing analysis

TECHNICAL SKILLS

Languages: Java, Python, C#, SQL, JavaScript, TypeScript, HTML/CSS

Developer Tools & Infra: Docker, Kubernetes, Terraform, AWS, Azure, IBM Cloud, CI/CD, GitHub Actions, Git, Prometheus, Splunk, PostgreSQL, MongoDB, Claude Code, Copilot

AI & Agentic Systems: LangChain, LangGraph, MCP, RAG, Prompt Engineering, Context Engineering

Frameworks & Libraries: React, Next.js, Node.js, Spring Boot, .NET, REST APIs